INCH-POUND

MIL-DTL-23413D(SH)
5 March 2026
SUPERSEDING
MIL-I-23413C(SH)
4 August 1981

DETAIL SPECIFICATION

INSERTS, WELDING, FILLER MATERIAL, COILED AND SOLID RINGS

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers consumable insert material in coil and solid ring forms for weld joints fabricated from carbon steel, alloy steels, aluminum alloy, and nickel-based alloys, including copper-nickel alloy (70/30).

1.2 Classification.

1.2.1 Classes and styles. Inserts are furnished in the following classes and styles, as specified (see 6.2):

a. Class 1 – Inverted T-shaped cross section (see figure 1).
   (1) Style A – Coiled insert filler material.
   (2) Style B – Preformed rings, open lap joint.
   (3) Style C – Preformed rings, open butt joint.

b. Class 2 – J-shaped cross section (see figure 1).
   (1) Style A – Coiled insert filler material.
   (2) Style B – Preformed rings, open lap joint.
   (3) Style C – Preformed rings, open butt joint.

c. Class 3 – Rectangular-shaped cross section (see figure 2).
   (1) Style D – Solid ring, 3/16-inch rim width.
   (2) Style E – Solid ring, 1/8-inch rim width.

d. Class 4 – Y-shaped cross section (see figure 3).
   (1) Style A – Coiled insert filler material.
   (2) Style C – Preformed rings, open butt joint.

e. Class 5 – Cross section is as specified (see 6.2).
   (1) Style A – Coiled insert filler material.
   (2) Style B – Preformed rings, open lap joint.
   (3) Style C – Preformed rings, open butt joint.
   (4) Style D – Solid rings.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

AMSC N/A FSC 3439

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

1.2.2 Types and sizes.

1.2.2.1 Types. Inserts are furnished in the types shown in table I, as specified (see 6.2).

TABLE I. Types.

Class	Insert type	Type designation identification	Alloy group identification
All	MIL-Ms-1	Ms-1	Group A (mild steel)
	MIL-Ms-2	Ms-2	
All	MIL-515 (1¼ Cr-½ Mo)	515	Group B (chromium-molybdenum steel)
	MIL-521 (2¼ Cr-1 Mo)	521	
	MIL-505 (5 Cr-½ Mo)	505	
All	MIL-308	308	Group C (austenitic chromium-nickel steels)
	MIL-308L	308L	
	MIL-310	310	
	MIL-312	312	
	MIL-316	316	
	MIL-316L	316L	
	MIL-348	348	
All	MIL-1100	1100	Group D (aluminum alloys)
	MIL-4043	4043	
	MIL-5356	5356	
All	MIL-52i	52i	Group E (nickel, nickel alloys, and copper-nickel alloys)
	MIL-60	60	
	MIL-61	61	
	MIL-62	62	
	MIL-67	67	
	MIL-6A	6A	
	MIL-82	82	
	MIL-82H	82H	
	MIL-625	625	

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

1.2.2.2 Sizes. Inserts are furnished in the sizes shown in table II, as specified (see 6.2).

TABLE II. Sizes.

Class and style	Figure	Size reference
Class 1 and 2, style A	Figure 1	3.2.2
Class 1 and 2, style B	Figure 1	3.2.3, table IX
Class 1 and 2, style C	Figure 1	3.2.4
Class 3, style D	Figure 2	3.3.2, table X
Class 3, style E	Figure 2	3.3.2, table XI
Class 4, style A	Figure 3	3.4.2
Class 4, style C	Figure 3	3.4.3
Class 5, style A	--	3.5.2
Class 5, style B	--	3.5.3, table IX
Class 5, style C	--	3.5.4
Class 5, style D	--	3.5.5

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-W-10430 - Welding Rods and Electrodes; Preparation for Delivery of

MIL-E-19933 - Electrodes and Rods – Welding, Bare, Chromium and Chromium-Nickel Steels

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of this publication are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). This publication is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables

(Copies of this publication are available online at https://asq.org.)

AMERICAN WELDING SOCIETY (AWS)

AWS A4.2M	-	Standard Procedures for Calibrating Magnetic Instruments to Measure the Delta Ferrite Content of Austenitic and Duplex Ferritic-Austenitic Stainless Steel Weld Metal
AWS A5.01M/A5.01	-	Welding and Brazing Consumables – Procurement of Filler Metals and Fluxes
AWS A5.30/A5.30M	-	Specification for Consumable Inserts

(Copies of these publications are available online at www.aws.org.)

ASTM INTERNATIONAL

ASTM A240/A240M	-	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

(Copies of this publication are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Materials. Insert materials shall be of uniform quality and condition and shall be free of piping, seams, segregations, and abrasions that would adversely affect the usability characteristics of the insert or properties of the fused material.

3.1.1.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.1.1.2 Material from previous production runs. Materials from previous production runs shall be excluded from any new production run to be offered for acceptance unless the materials have been determined to be acceptable for meeting all requirements of this specification (see 4.5).

3.1.1.3 Asbestos. The inserts and their containers shall be asbestos-free (see 4.4.4). The containers shall be marked “Asbestos-Free” (see 6.2).

3.1.2 Alloy identity.

3.1.2.1 All phases of production prior to final marking. The material shall be of the type specified when tested in accordance with 4.3.4 during all phases of production prior to final marking.

3.1.2.2 After final marking. When specified (see 6.2), each cut length of material in preformed insert rings and each end of each spool or coil of insert material shall be of the material type specified when tested for alloy identity in accordance with 4.3.4 after final marking. Identification marking of each ring, spool, or coil tested after final marking and meeting the acceptance criteria of 4.3.4 shall include the words “ALLOY TESTED”.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

3.1.3 Finish. Inserts shall have smoothly finished surfaces and shall be free from slivers, depressions, scratches, and scale or any foreign matter that would adversely affect the usability characteristics of the insert or properties of the fused metal.

3.1.3.1 Rust preventatives. When specified (see 6.2), rust preventatives may be used as protective coating if they are readily removable with solvents by wiping or dipping, and when a rust preventative has been used, cleaning procedures for removing the rust preventative are clearly indicated on the packaging.

3.1.3.2 Special requirements for group C and group E inserts.

3.1.3.2.1 Grit or vapor blasting. Group C and group E inserts shall not be grit or vapor blasted.

3.1.3.2.2 Bright annealing. When specified (see 6.2), group C and group E inserts shall be bright annealed. When specified (see 6.2), MIL-52i, MIL-62, MIL-82, MIL-82H, and MIL-625 inserts shall be special bright hydrogen annealed at 1,800±50 °F in dry hydrogen having an entrance dew point of minus 80 °F or lower. Following bright annealing or special bright hydrogen annealing, inserts shall not be subjected to any surface treatment other than solvent cleaning.

3.1.4 Chemical composition. The chemical compositions of inserts shall be as specified in tables III through VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

TABLE III. Chemical composition – group A (mild steels). [1/]

Type	Carbon (percent)	Manganese (percent)	Phosphorus (percent)	Sulfur (percent)	Silicon (percent)	Aluminum [2/] (percent)	Zirconium [2/] (percent)	Titanium [2/] (percent)
MIL-Ms-1	0.06	1.00–1.40	0.025	0.03	0.40–0.60	0.05–0.15	0.02–0.12	0.05–0.15
MIL-Ms-2	0.20	0.90–1.30	0.025	0.03	0.45–0.65	--	--	--

NOTES:

1/ Single values are maximum percentages, except where otherwise specified.

2/ Al + Zr + Ti = 0.15 minimum.

TABLE IV. Chemical composition – group B (chromium-molybdenum steels). [1/]

Type	Carbon (percent)	Manganese (percent)	Phosphorus (percent)	Sulfur (percent)	Silicon (percent)	Aluminum (percent)	Chromium (percent)	Molybdenum (percent)	Copper (percent)	Nickel (percent)
MIL-515	0.10	0.40–0.70	0.03	0.03	0.40–0.70	0.15	1.2–1.5	0.40–0.65	0.15	--
MIL-521	0.10	0.40–0.70	0.03	0.03	0.25–0.50	0.15	2.3–2.7	0.90–1.1	0.15	0.40
MIL-505	0.10	0.40–0.75	0.025	0.025	0.25–0.50	0.15	5.0–6.0	0.45–0.65	--	--

NOTE:

1/ Single values are maximum percentages, except where otherwise specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

TABLE V. Chemical composition – group C (austenitic chromium-nickel steels). 1/

Type	Carbon (percent)	Chromium (percent)	Nickel (percent)	Molybdenum (percent)	Niobium plus tantalum (percent)	Manganese (percent)	Silicon (percent)	Phosphorus (percent)	Sulfur (percent)	Cobalt (percent)
MIL-308 2/	0.08	19.5–22.0	9.0–11.0	--	--	1.0–2.5	0.25–0.60	0.03	0.03	0.10
MIL-308L 2/	0.03	19.5–22.0	9.0–11.0	--	--	1.0–2.5	0.25–0.60	0.03	0.03	0.10
MIL-310	0.08–0.15	25.0–28.0	20.0–22.5	--	--	1.0–2.5	0.25–0.60	0.03	0.03	0.10
MIL-312	0.08–0.15	28.0–32.0	8.0–10.5	--	--	1.0–2.5	0.25–0.60	0.03	0.03	0.10
MIL-316	0.08	18.0–20.0	11.0–14.0	2.0–3.0	--	1.0–2.5	0.25–0.60	0.03	0.03	0.10
MIL-316L	0.03	18.0–20.0	11.0–14.0	2.0–3.0	--	1.0–2.5	0.25–0.60	0.03	0.03	0.10
MIL-348 2/	0.08	19.0–21.5	9.0–11.0	--	10 x C min. 3/ to 1.0 max.	1.0–2.5	0.25–0.60	0.03	0.03	0.10

NOTES:

1/ Single values shown are maximum percentages, except where otherwise specified.

2/ Delta ferrite content of weld metal (fused insert material) shall be from 4 to 10 ferrite number (FN) determined by the Magne-Gage, or equal (see figure 5).

3/ Tantalum content shall not exceed 0.10 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

TABLE VI. Chemical composition – group D (aluminum alloys). [1/]

Type [2/]	Copper (percent)	Magnesium (percent)	Manganese (percent)	Chromium (percent)	Silicon (percent)	Iron (percent)	Zinc (percent)	Titanium (percent)	Beryllium (percent)	Other		Aluminum (percent)
										Each (percent)	Total (percent)	
MIL-1100	0.02	--	0.05	--	[3/]	[3/]	0.10	--	0.0008	0.05	0.15	99.0–99.5
MIL-4043	0.30	0.05	0.05	--	4.5–6.0	0.8	0.10	0.20	0.0008	0.05	0.15	Remainder
MIL-5356	0.10	4.5–5.5	0.05–0.20	0.05–0.20	[4/]	[4/]	0.10	0.06–0.20	0.0008	0.05	0.15	Remainder

NOTES:

[1/] Single values shown are maximum percentages, except where otherwise specified.

[2/] The alloy designations are in accordance with the system approved by the Aluminum Association.

[3/] Iron plus silicon = 1.0 percent maximum.

[4/] Iron plus silicon = 0.50 percent maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

TABLE VII. Chemical composition – group E (nickel, nickel alloys, and copper-nickel alloys). [1/] [2/]

Type	Carbon [3/] (percent)	Manganese (percent)	Iron (percent)	Sulfur (percent)	Phosphorus (percent)	Silicon (percent)	Copper (percent)	Nickel plus cobalt (percent)	Aluminum (percent)	Titanium (percent)	Chromium (percent)	Niobium plus tantalum (percent)	Molybdenum (percent)	Other elements total [4/] (percent)
MIL-52i	0.03–0.05	2.50–3.50	2.00–3.00	0.002	0.008	0.01–0.10	0.10	Balance [5/]	0.20	0.20–0.40	26.0–28.0	2.20–2.40 [6/]	0.010	0.50 [7/]
MIL-60	0.15	4.00	2.50	0.02	0.020	1.50	Balance	62.0–69.0	1.25	1.5–3.0	--	--	--	0.50
MIL-61	0.15	1.00	1.00	0.01	0.030	0.75	0.25	93.0 min.	1.50	2.0–3.5	--	--	--	0.50
MIL-62	0.08	1.00	6.0–8.0	0.015	0.030	0.35	0.50	70.0 [5/] min.	--	--	14.0–17.0	1.5–3.0 [6/]	--	1.0
MIL-67	0.04	1.00	0.4–0.7	0.01	0.020	0.15	Balance	29.0–32.0	--	0.2–0.5	--	--	--	0.50
MIL-6A	0.08	2.00–2.75	8.00	0.015	0.030	0.35	0.50	67.0 [5/] min.	--	2.5–3.5	14.0–17.0	--	--	0.50
MIL-82	0.10	2.50–3.50	3.00	0.015 [8/]	0.030	0.50	0.50	67.0 [5/] min.	--	0.75	18.0–22.0	2.0–3.0 [6/]	--	0.50
MIL-82H	0.030–0.10	2.50–3.50	3.00	0.015 [8/]	0.030	0.50	0.50	67.0 [5/] min.	--	0.75	18.0–22.0	2.0–3.0 [6/]	--	0.50
MIL-625	0.10	0.50	5.00	0.015 [8/]	0.20	0.50	1.0	58.0 [5/] min.	0.40	0.40	20.0–23.0	3.14–4.15 [6/]	8.0–10.0	0.50 [9/]

NOTES:

1/ Analysis shall be made for the elements for which specific values are shown. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine that the total of these other elements is not present in excess of the limits specified for "Other elements, total".

2/ Single values shown are maximum percentages, except where otherwise specified.

3/ When specified (see 6.2), a minimum carbon content shall be required.

4/ Includes lead, tin, and zinc.

5/ Cobalt shall not exceed 0.10 percent.

6/ Tantalum shall not exceed 0.30 percent.

7/ Nitrogen 0.015–0.025 percent. Magnesium 0.001–0.010 percent. No intentional additions of unspecified elements.

8/ When specified (see 6.2), sulfur shall not exceed 0.007 percent.

9/ Lead shall not exceed 0.010 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

3.2 Requirements – classes 1 and 2.

3.2.1 Shapes. The shape or configuration of cross sections shall be as shown on figure 1.

3.2.1.1 Dimensions and tolerances. The dimensions and tolerances of the cross sections shall be as specified in tables VIII and IX.

TABLE VIII. Class 1, 2, and 4 inserts – cross-sectional dimensions and tolerances.

Insert size	Legend (figures 1 and 3)	Class 1 Inverted T-shaped 1/		Class 2 J-shaped 1/		Class 4 Y-shaped 2/	
		Dimensions (inch)	Tolerances (inch)	Dimensions (inch)	Tolerances (inch)	Dimensions (inch)	Tolerances (inch)
1/8 inch	D	0.125	±0.004	0.086	+0.011 -0.005	0.165	±0.010
	W	0.047	+0.002 -0.012	0.047	+0.002 -0.012	0.078	±0.010
	H	0.055	+0.012 -0.002	0.055	+0.012 -0.002	0.140	±0.010
	H/2	--	--	--	--	0.072	±0.010
5/32 inch	D	0.156	±0.005	0.110	+0.012 -0.010	0.205	±0.015
	W	0.063	+0.003 -0.014	0.063	+0.003 -0.014	0.093	±0.015
	H	0.063	+0.014 -0.003	0.063	+0.014 -0.010	0.175	±0.010
	H/2	--	--	--	--	0.093	±0.010

NOTES:

1/ The offset between the center of the rib (W) and the center of the land (D) shall not exceed 0.005 inch.

2/ Additional dimensions and tolerances – class 4 Y-shaped inserts (see figure 3):

		1/8-inch size	5/32-inch size
(a)	Radius of rib	0.044±0.005	0.050±0.005
(b)	Angle between side of rib and center line	1 to 2 degrees	1 to 2 degrees
(c)	Rib offset	±0.010	±0.015
(d)	Height of rib along center line	0.100±0.010	0.115±0.010
(e)	Angle between top surface of inclined arm and center line	50±5 degrees	50±5 degrees
(f)	Length of inclined arm	0.085±0.010	0.125±0.010

From a practical inspection standpoint, the D, W, H, and H/2 dimensions have the most to do with usability and weldability. The (a) through (f) dimensions are furnished to complete the description of the cross-sectional configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

TABLE IX. Classes 1, 2, and 5, style B – minimum overlap.

Nominal pipe size (inches)	Circumferential overlap (inch)
½	0.250
¾	0.250
1	0.250
1¼	0.250
1½	0.250
2	0.250
2½	0.250
3	0.375
3½	0.375
4	0.375
5	0.500
6	0.500
8	0.625
10	0.750
12	0.750
14	0.750

3.2.2 Style A – coil details. Unless otherwise specified (see 6.2), coiled inserts shall be furnished in 50-foot lengths wound on a 12-inch rim. The cast of the insert shall be 9 inches nominal diameter.

3.2.3 Style B – dimensions. Preformed rings with open lap joints shall be made to order based on a nominal pipe size specified by the contracting activity (see 6.2). Circumferential overlap shall be as specified in table IX. For nominal pipe sizes greater than 14 inches diameter, circumferential overlap shall be as specified (see 6.2).

3.2.4 Style C – dimensions. Preformed rings with open butt joints shall be made to order and fitted to a template that will be furnished by the contracting activity (see 6.2).

3.2.5 Identification. The type designation identification (see table I) shall be imprinted or stamped at 6 inch intervals on style A (coiled) inserts, and shall appear at least once on each style B and style C preformed insert at the location shown on figure 1. Imprints shall be legible, resist effacement by contact incidental to normal handling, and shall not interfere with insert usability. The depth of the imprint shall not exceed 0.015 inch.

3.2.6 Fused insert welds. Class 1 and 2 inserts shall demonstrate good weldability when sampled in accordance with 4.1.1.4 and tested in accordance with 4.3.2.

3.3 Requirements – class 3.

3.3.1 Design. Unless otherwise specified (see 6.2), solid insert rings shall be of the design shown on figure 2.

3.3.2 Dimensions and tolerances. Solid rings shall be furnished in accordance with the dimensions (styles) shown on figure 2 and the inside diameter (I.D.) and outside diameter (O.D.) dimensions specified in tables X and XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

TABLE X. Class 3, style D inserts – dimensions.

Pipe dimensions			Ring diameter [2/]	
Nominal diameter (inches)	**Schedule number (see 6.2) [1/]**	**I.D. [2/] (inches)**	**Ring O.D. for nominal pipe diameter (inches)**	**Ring I.D. for nominal pipe diameter (inches)**
2	10S	2.157	2.43	2.06
	40	2.067	2.34	1.97
	80	1.939	2.22	1.85
2½	10S	2.635	2.91	2.54
	40	2.469	2.75	2.38
	80	2.323	2.60	2.23
3	10S	3.260	3.54	3.17
	40	3.068	3.35	2.98
	80	2.900	3.18	2.81
3½	10S	3.760	4.04	3.67
	40	3.548	3.82	3.45
	80	3.364	3.64	3.27
4	10S	4.260	4.54	4.17
	40	4.026	4.30	3.93
	80	3.826	4.10	3.73
5	5S	5.345	5.62	5.25
	10S	5.295	5.57	5.20
	40	5.047	5.32	4.95
	80	4.813	5.09	4.72
6	5S	6.407	6.68	6.31
	10S	6.357	6.63	6.26
	40	6.065	6.34	5.97
	80	5.761	6.04	5.67
8	5S	8.407	8.68	8.31
	10S	8.329	8.61	8.24
	40	7.981	8.26	7.89
	80	7.625	7.90	7.53
10	5S	10.482	10.76	10.39
	10S	10.420	10.70	10.33
	40	10.020	10.30	9.93
	80S	9.750	10.03	9.66
	80	9.564	9.84	9.47

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

TABLE X. Class 3, style D inserts – dimensions – Continued.

Pipe dimensions			Ring diameters [2/]	
Nominal diameter (inches)	**Schedule number (see 6.2) [1/]**	**I.D. [2/] (inches)**	**Ring O.D. for nominal pipe diameter (inches)**	**Ring I.D. for nominal pipe diameter (inches)**
12	5S	12.420	12.70	12.33
	10S	12.390	12.67	12.30
	40S	12.000	12.28	11.91
	40	11.938	12.22	11.85
	80S	11.750	12.03	11.66
	80	11.376	11.65	11.28
14	10	13.500	13.78	13.41
	40	13.126	13.40	13.03
	80	12.500	12.78	12.41
16	10	15.500	15.78	15.41
	40	15.000	15.28	14.91
	80	14.314	14.59	14.22

NOTES:

1/ Schedule number followed by “S” applies to group C types.

2/ Class 3, style D insert rings are normally furnished to match nominal internal diameters of the pipe. The dimensions of such rings are determined as follows:

Ring I.D. = Pipe I.D. - 0.094 inch

Ring O.D. = Ring I.D. + 0.375 inch

Where special applications demand rings having ⅛ inch instead of the 3/16 inch dimension shown on figure 2, the narrower width (style E) shall be made available as shown in table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

TABLE XI. Class 3, style E inserts – dimensions.

Pipe dimensions			Ring diameters [2/]	
Nominal diameter (inches)	**Schedule number (see 6.2) [1/]**	**I.D. [2/] (inches)**	**Ring O.D. for nominal pipe diameter (inches)**	**Ring I.D. for nominal pipe diameter (inches)**
¼	10S	0.410	0.57	0.32
	40	0.364	0.52	0.27
	80	0.302	0.46	0.21
⅜	10S	0.545	0.70	0.45
	40	0.493	0.65	0.40
	80	0.423	0.58	0.33
½	5S	0.710	0.87	0.62
	10S	0.674	0.83	0.58
	40	0.622	0.78	0.53
	80	0.546	0.70	0.45
¾	5S	0.920	1.08	0.83
	10S	0.884	1.04	0.79
	40	0.824	0.98	0.73
	80	0.742	0.90	0.65
1	5S	1.185	1.34	1.09
	10S	1.097	1.25	1.00
	40	1.049	1.21	0.96
	80	0.957	1.11	0.86
1¼	5S	1.530	1.69	1.44
	10S	1.442	1.60	1.35
	40	1.380	1.54	1.29
	80	1.278	1.43	1.18
1½	5S	1.770	1.93	1.68
	10S	1.682	1.84	1.59
	40	1.610	1.77	1.52
	80	1.500	1.66	1.41
2	5S	2.245	2.40	2.15
	10S	2.157	2.31	2.06
	40	2.067	2.22	1.97
	80	1.939	2.10	1.85
2½	5S	2.709	2.87	2.62
3	5S	3.334	3.49	3.24
3½	5S	3.834	3.99	3.74
4	5S	4.334	4.49	4.24

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

TABLE XI. Class 3, style E inserts – dimensions – Continued.

NOTES:

1/ Schedule number followed by "S" applies to group C types.

2/ Special ring sizes (non-standard) shall be obtainable when specified (see 6.2). The dimensions of such rings are determined as follows:

Ring I.D. = Pipe I.D. - 0.094 inch

Ring O.D. = Ring I.D. + 0.250 inch

3.3.3 Flatness. Solid insert rings shall be flat to within 0.015 inch at any location when measured on a surface plate using feeler gages.

3.3.4 Identification. The type designation identification (see table I) and applicable pipe schedule number (see table X, table XI, and style D or E, as applicable) shall be imprinted or stamped on each ring (see figure 2). Imprints shall be legible, resist effacement during normal handling, and shall not interfere with insert usability. The depth of the imprint shall not exceed 0.015 inch.

3.4 Requirements – class 4.

3.4.1 Shapes. The shape or configuration of cross sections shall be as shown on figure 3.

3.4.1.1 Dimensions and tolerances. The dimensions and tolerances of the cross sections shall be as specified in table VIII.

3.4.2 Style A – coil details. Unless otherwise specified (see 6.2), coiled inserts shall be furnished in 50-foot lengths wound on a 12-inch rim. The cast of the insert shall be 9 inches nominal diameter.

3.4.3 Style C – dimensions. Preformed rings with open butt joints shall be made to order and fitted to a template that will be furnished by the contracting activity (see 6.2).

3.4.4 Identification. The type designation number shall be imprinted or stamped at 6-inch intervals on style A (coiled) inserts and shall appear at least once on each style C preformed insert, which shall additionally bear the applicable nominal pipe size. Imprints shall be legible, resist effacement by contact incidental to normal handling, and shall not interfere with insert usability. Stamped designations shall have a depth equal to or greater than a Rockwell C hardness impression on the material being marked. It is recommended that, as an alternate identification for all inserts, but especially those excessively hardened from finish rolling or those too small for satisfactory imprinting, a "flag-tag" type marker be used for each circular unit of all styles.

3.5 Requirements – class 5.

3.5.1 Shapes. The shape or configuration of cross sections shall be as specified (see 6.2).

3.5.1.1 Dimensions and tolerances. The dimensions and tolerances of the cross sections shall be as specified (see 6.2).

3.5.2 Style A – coil details. Unless otherwise specified (see 6.2), coiled inserts shall be furnished in 50-foot lengths wound on a 12-inch rim. The cast of the insert shall be 9 inches nominal diameter.

3.5.3 Style B – dimensions. Preformed rings with open lap joints shall be made to order based on a nominal pipe size specified by the contracting activity (see 6.2). Circumferential overlap shall be as specified in table IX. For nominal pipe sizes greater than 14 inches diameter, circumferential overlap shall be as specified (see 6.2).

3.5.4 Style C – dimensions. Preformed rings with open butt joints shall be made to order and fitted to a template that will be furnished by the contracting activity (see 6.2).

3.5.5 Style D. The I.D. and O.D. dimensions and tolerances shall be as specified (see 6.2).

3.5.5.1 Flatness. Solid insert rings shall be flat to within 0.015 inch at all locations when measured on a surface plate using feeler gages.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

3.5.6 Identification. The type designation identification (see table I) shall be imprinted or stamped at 6-inch intervals on style A (coiled) inserts and shall appear at least once on each style B and style C preformed insert. The type designation identification (see table I) and pipe schedule number (if applicable; see tables X and XI) shall be imprinted or stamped on each style D (solid ring) insert. Imprints shall be legible, resist effacement during normal handling, and shall not interfere with insert usability. The depth of the imprint shall not exceed 0.015 inch.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include the examination of 4.2 and the tests of 4.3 through 4.5, as specified in table XII. Sampling shall be as specified in 4.1.1 through 4.1.1.4.

TABLE XII. Summary of conformance inspections.

Test or examination	Sampling paragraph	Verification paragraph	Requirement paragraph
Pre-production inspection of equipment	--	4.5	3.1.2.1
Alloy identity test	--	4.3.4	3.1.2
Visual and dimensional examination	4.1.1.3	4.2	3.1.3, 3.2 through 3.5
Chemical analyses	4.1.1.4	4.3.1	3.1.4 and tables III through VII
Fusion test	4.1.1.4	4.3.2	3.2.6
Delta ferrite weld test (for some group C inserts)	4.1.1.4	4.3.3	Table V
Packaging and marking	4.1.1.2	4.3.5	(see 6.2)
Asbestos-free	4.1.1.2	4.4.4	3.1.1.3

4.1.1 Sampling for conformance inspection.

4.1.1.1 Lot. For the purpose of selecting samples for conformance inspection, a lot of coiled or solid inserts shall be as defined for each test as specified in 4.1.1.2 through 4.1.1.4.1.

4.1.1.2 Sampling for packaging examinations. Unless otherwise specified (see 6.2), sampling of each lot for verification of conformance to the packaging requirements of the contract shall be in accordance with MIL-W-10430 (see 3.1.1.3, 4.3.5, and 4.4.4).

4.1.1.3 Sampling for visual and dimensional examination of inserts.

4.1.1.3.1 Classes 1, 2, 4, and 5 – style A. A lot shall be the number of coils having the same cross section (see 1.2.1) produced from the same heat of metal and offered for delivery at the same time. Sample coils shall be selected from each lot in accordance with ANSI/ASQ Z1.4 at inspection level III and shall be examined as specified in 4.2. The maximum allowable number of defects shall be as specified (see 6.2).

4.1.1.3.2 Classes 1 and 2, styles B and C, all class 3 and class 4, style C, and class 5, styles B, C, and D. A lot shall be the number of preformed ring inserts having the same cross section (see 1.2.1) produced from the same heat of metal and offered for delivery at the same time. Sample inserts shall be selected from each lot in accordance with ANSI/ASQ Z1.4 at inspection level I. The sample inserts shall be selected in equal number from not less than five unit containers and examined as specified in 4.2. The maximum allowable number of defects shall be as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

4.1.1.4 Sampling for chemical analysis, fusion test, and delta ferrite weld test (all classes). A lot shall be the number of coils, preformed rings, or solid rings of insert material having the same cross section (see 1.2.1) produced from the same heat of material and offered for delivery at the same time. One of each type of test shall be conducted per 4.3 for each lot of material. For preformed rings or solid rings, the starting coils (i.e., the same cross section and heat) used to produce rings may be tested in lieu of testing material from finished rings. Failure of any coil or ring to meet the requirements of this specification shall be cause for rejection of the lot.

4.1.1.4.1 Test waiver. When specified by the purchasing activity (see 6.2), chemical analysis, fusion testing, and delta ferrite weld testing may be waived on a shipment made from the same heat as a previous shipment that was tested and determined to comply with the requirements herein.

4.2 Visual and dimensional examination. Sample inserts selected as specified in 4.1.1.3.1 and 4.1.1.3.2 shall be examined and measured to verify conformance to all of the requirements of this specification that do not involve tests. Each coil selected in accordance with 4.1.1.3.1 shall be examined and measured at nine points distributed uniformly throughout its length.

4.3 Test procedures.

4.3.1 Chemical analysis. Samples selected in accordance with 4.1.1.4 shall be analyzed for chemical composition by any accepted analytical method agreed upon by the contractor and the purchaser as specified (see 6.2 and 6.6).

4.3.2 Fused insert welds. Fused insert welds shall be made as specified on figure 4 and notes applying thereto. Fused insert welds shall be inspected as follows.

4.3.2.1 Fused insert contour. The contour of the backside surface for the fused insert shall comply with the following requirements. Any nonconformance shall be cause for rejection.

a. Reinforcement shall not exceed 0.031 inch.

b. Concavity shall not exceed 0.031 inch.

c. The contour shall have a uniform radius and shall blend smoothly into the base material.

4.3.2.2 Linear indications. Upon completion of fusion, the weld joint shall be visually examined for compliance with 4.3.2.1 and shall be liquid penetrant inspected in accordance with T9074-AS-GIB-010/271 with non-water washable penetrant. The fused insert shall be free of linear indications as revealed by the liquid penetrant inspection.

4.3.2.3 Internal weld surface. An internal weld surface of the fused insert showing oxide scale accompanied by a wrinkled or crystalline surface appearance shall be cause for rejection. However, tightly adhering, iridescent temper films shall be considered acceptable.

4.3.3 Delta ferrite weld test. The delta ferrite weld test shall be conducted in accordance with figure 5 and notes thereto for each group C insert type required by table V. The delta ferrite of weld metal (fused insert material) shall be determined by the Magne-Gage, or equal, calibrated using AWS A4.2M and reported as FN. When specified (see 6.2), for group C inserts manufactured from the same heat of material as filler material in accordance with MIL-E-19933, delta ferrite weld test results in accordance with MIL-E-19933 may be used in lieu of the test required by this section.

4.3.4 Alloy identity. Alloy identity methods and procedures shall be in accordance with AWS A5.01M/A5.01 Department of Defense (DoD) supplementary requirements (see 3.1.2). The test method(s) and procedure(s) shall be submitted for approval to the Technical Authority (see 6.4).

4.3.5 Inspection of marking and packaging. Packaging and marking shall be inspected for compliance with the contract (see 5.1, 6.2, and 6.7).

4.4 Certificate of conformance (CoC). Unless otherwise specified (see 6.2), a CoC shall be furnished with each lot of material offered for acceptance. The minimum certification test data shall be in accordance with 4.4.1 through 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

4.4.1 Test results. The CoC shall include quantitative results for the specified chemical composition tests herein (see 4.3.1). Delta ferrite content shall be reported in FN (see 4.3.3). The CoC shall also include qualitative results of nondestructive tests and other inspections and tests. The cause for any retest shall be reported for information and the test results that failed to meet those specified requirements shall be reported.

4.4.2 Welding information. Welding processes and operational parameters for fusibility test assemblies shall also be reported, including results of fused insert welds (see 4.3.2).

4.4.3 Other information. The manufacturer or distributor name and address, customer, customer order number, specification, heat and lot, class and style, size, insert type, and finish shall be identified.

4.4.4 Certification statement. The CoC shall state that each lot has been sampled, tested, and inspected in accordance with the specification requirements to produce a consumable insert that meets the requirements, and that the inserts are asbestos-free. The CoC shall be signed by a responsible representative of the manufacturer.

4.4.5 Test certificates. Where test certificates issued by the manufacturer contain the above information (see 4.4.1 through 4.4.4), a separate CoC will not be required.

4.5 Pre-production inspection of equipment. Prior to the start of a new production run, where a different insert type is to be produced, inspection shall be performed to ensure that equipment and process lines are purged of material from the previous production run that does not meet the requirements herein. This inspection shall be performed for all equipment used in manufacturing operations where the material is not segregated and positively identified (i.e., cutting to length, extruding, and baking) (see 3.1.2.1).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Coiled insert material or solid rings are intended for use in making sound, smooth, uniform, crack-free root welds where accessibility is limited to one side. Insert alloy composition and the composition of subsequently deposited weld metal should be compatible with base material. This specification contains requirements above and beyond those of commercial specifications for use in critical military applications. Current stocks of consumable insert types specified in this document may be used until depleted; future requisitions for consumable insert types specified in this specification should refer to the current version of this specification.

6.1.1 Class 4, style A – coil details. Class 4 inserts tend to twist when the rim is coiled. This is a result of an essentially triangular configuration, narrower at inner rim contact point than at the outer surface. However, special coiling conditions may be specified (see 6.2).

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class, style, type, and size (for all classes) (see 1.2.1, 1.2.2.1, and 1.2.2.2).

c. Shape, configuration, dimensions, and tolerances of cross sections of class 5 inserts (see 1.2.1, 3.5.1, and 3.5.1.1).

d. Requirement for marking on the package indicating that packaging is asbestos-free (see 3.1.1.3).

e. Whether alloy identity testing of each cut length of wire in preformed inserts and each end of each spool or coil of insert wire after final marking is required (see 3.1.2.2).

f. When rust preventative is required for group A and B inserts (see 3.1.3.1).

g. Whether bright annealing of group C and E inserts is required (see 3.1.3.2.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

h. Whether special bright hydrogen annealing of MIL-52i, MIL-62, MIL-82, MIL-82H, and MIL-625 inserts is required (see 3.1.3.2.2).

i. Whether minimum carbon content for group E inserts is required (see table VII, note 2).

j. Whether sulfur for group E inserts is not to exceed 0.007 percent (see table VII, note 8).

k. If style A coiled inserts are to be furnished in other than 50-foot lengths wound on a 12-inch rim (see 3.2.2, 3.4.2, and 3.5.2).

l. Nominal pipe size for all orders for style B of classes 1, 2, and 5 and templates to be furnished for all orders for style C of classes 1, 2, 4, and 5 (see 3.2.3, 3.2.4, 3.4.3, 3.5.3, and 3.5.4).

m. Circumferential overlap of class 1, 2, or 5 style B inserts of greater than 14 inches nominal pipe size (see 3.2.3 and 3.5.3).

n. Whether class 3 solid insert rings are to be other than as specified on figure 2 (see 3.3.1).

o. Nominal pipe size and schedule number for class 3 (see tables X and XI) and, if applicable, class 5, style D.

p. Whether special ring sizes are required (see table XI, note 2).

q. I.D. and O.D. dimensions and tolerances for class 5, style D inserts (see 3.5.5).

r. If sampling for packaging requirement verification should be other than MIL-S-10430 (see 4.1.1.2).

s. Maximum allowable number of defects (see 4.1.1.3.1, 4.1.1.3.2, and 6.5).

t. If a waiver is authorized for shipments made from the same heat as a previous shipment that was tested and determined to meet requirements (see 4.1.1.4.1).

u. Accepted analytical method agreed upon by the contractor and the purchaser (see 4.3.1 and 6.6).

v. If delta ferrite weld test results from the same heat of material used for MIL-E-19933 filler material may be substituted for delta ferrite weld test results for group C inserts (see 4.3.3).

w. When a CoC is not required (see 4.3.5).

x. Packaging requirements and additional recommendations (see 5.1 and 6.7).

y. Special coiling requirements for class 4, style A, if required (see 6.1.1).

z. Use of polystyrene (loose-fill) material (see 6.7.1).

aa. Responsibility for inspection (see 6.8).

bb. Responsibility for compliance (see 6.9).

cc. Inspection system requirements (see 6.10).

6.3 Inspection after delivery. Post delivery inspection of inserts to determine conformance to this specification and for acceptance thereof is the responsibility of the purchaser. When receipt inspection testing types MIL-308, MIL-308L, and MIL-348, the purchaser may consider the following delta ferrite test result limits acceptable:

a. A maximum of 14 FN.

b. An average minimum level of 3 FN with no individual test result less than 2 FN.

Vendor test results should meet the specified 4 to 10 FN.

6.4 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.5 Recommendations for percent defective level. It is recommended that an acceptable quality level equal to 1.5 percent defective be utilized for determining the acceptance levels when applying the procedures of ANSI/ASQ Z1.4. This is the same value specified in the previous revisions of this specification (see 4.1.1.3.1, 4.1.1.3.2, and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

6.6 Chemical analysis. Chemical analysis may be in accordance with the following specifications, or any acceptable analytical method as agreed upon between the contractor and purchaser (see 6.2):

a. Group A – ASTM E350.

b. Group B – ASTM E350 or ASTM E352.

c. Group C – ASTM E353.

d. Group D – ASTM E1251 or ASTM E3061.

e. Group E – ASTM E354 or ASTM E1473.

6.6.1 Disputes. In the event of a dispute, the appropriate wet chemical analysis of the specimen should be used.

6.7 Additional packing and marking recommendations. Unless otherwise specified (see 6.2), packaging and marking of packaging should be in accordance with AWS A5.30/A5.30M. Packing should be marked with the insert type, group, and class, as applicable, in accordance with AWS A5.30/A5.30M. Packing should also be marked when rust preventative is used (see 3.1.3.1).

6.7.1 Use of polystyrene (loose-fill) material for domestic shipment and early use. Unless otherwise specified (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early use for packaging and packing applications such as cushioning, filler, and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) should be marked and labeled as follows:

"CAUTION

Contents cushioned etc., with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard,
or transparent flexible cellular material."

6.8 Responsibility for inspection recommendations. Unless otherwise specified (6.2), the contractor should be responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Unless otherwise specified (see 6.2), the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the purchaser. The purchaser reserves the right to perform any of the inspections set forth herein where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

6.9 Responsibility for compliance recommendations. The inspections set forth in this specification should become part of the contractor's overall inspection system or quality program (see 6.10). The absence of any inspection requirements in the specification should not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the purchaser for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the purchaser to accept defective material (see 6.2).

6.10 Inspection system requirement recommendations. The contractor and their subcontractors should provide and maintain an inspection system acceptable to the Technical Authority for the supplies or services covered by the contract or order. The inspection system should be in accordance with one of the following, unless an equivalent system is approved (see 6.2):

a. ISO 9003 (ANSI/ISO/ASQ 09003) with STR-ISO 9000 Supplement C. Supplements B or A are also acceptable.

b. ISO 9002 (ANSI/ISO/ASQ 09002) with STR-ISO 9000 Supplement C. Supplements B or A are also acceptable.

c. ISO 9001 (ANSI/ISO/ASQ 09001) with STR-ISO 9000 Supplement C. Supplements B or A are also acceptable.

d. MIL-I-45208.

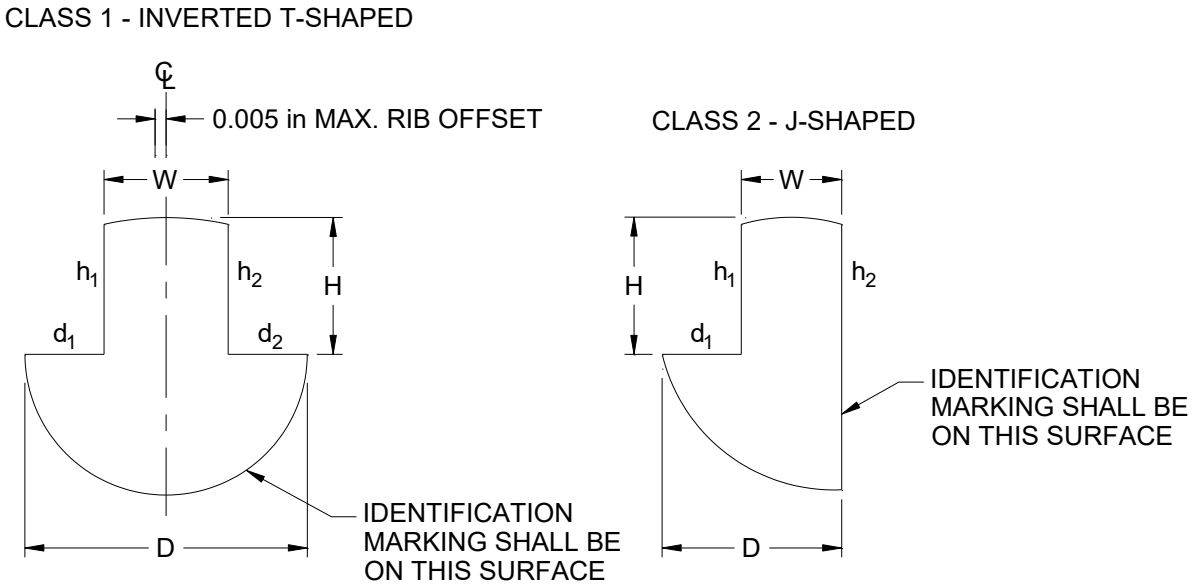
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

6.11 CoC considerations. The CoC, as required in this specification, is an essential element for ensuring that the material being purchased is in conformance with the specification. The certificate content, as required herein, provides valuable information regarding the welding parameters used for the required tests, expected performance, actual weld metal chemistry, whether the product was subjected to a stress relief heat treatment, information regarding test failures, and other information that links the certificate to the manufacturer, purchaser, specification, lot, etc. Navy technical and safety program documentation typically requires certain information included in the prescribed CoC for use as objective quality evidence demonstrating specification conformance and compliance as well as traceability for materials used within prescribed ship systems. Such documentation includes, but is not limited to, SUBSAFE; Level I; Deep Submergence Scope of Certification (DSSOC); and ship hull, machinery, and piping systems fabrication documents. A CoC may also be required by customs. The CoC also provides beneficial information in support of hull structure and component failure analyses; tracking problematic heats or lots of material throughout the supply system or onboard ships; and material, component, and specification improvements. These added benefits are possible because of the unique tracking information provided within the CoC and the unique set of mechanical properties and chemical content provided for each lot of material.

6.12 Subject term (key word) listing.

Carbon steel
Consumable insert
Copper alloy
Nickel alloy
Stainless steel

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



NOTES:

1. Lands (d_1, d_2) on either side of the rib shall be on the same plane within 0.005 inch.
2. Rib surfaces (h_1, h_2) shall be parallel within 0.002 inch and square with lands (d_1, d_2) within 0.005 inch.

FIGURE 1. Class 1 and class 2 inserts – cross-sectional configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

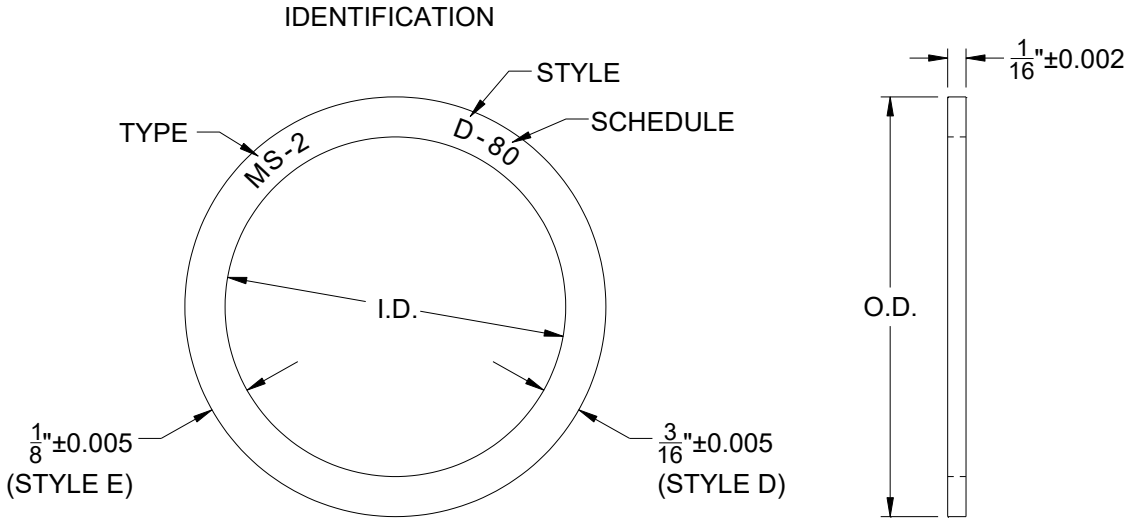
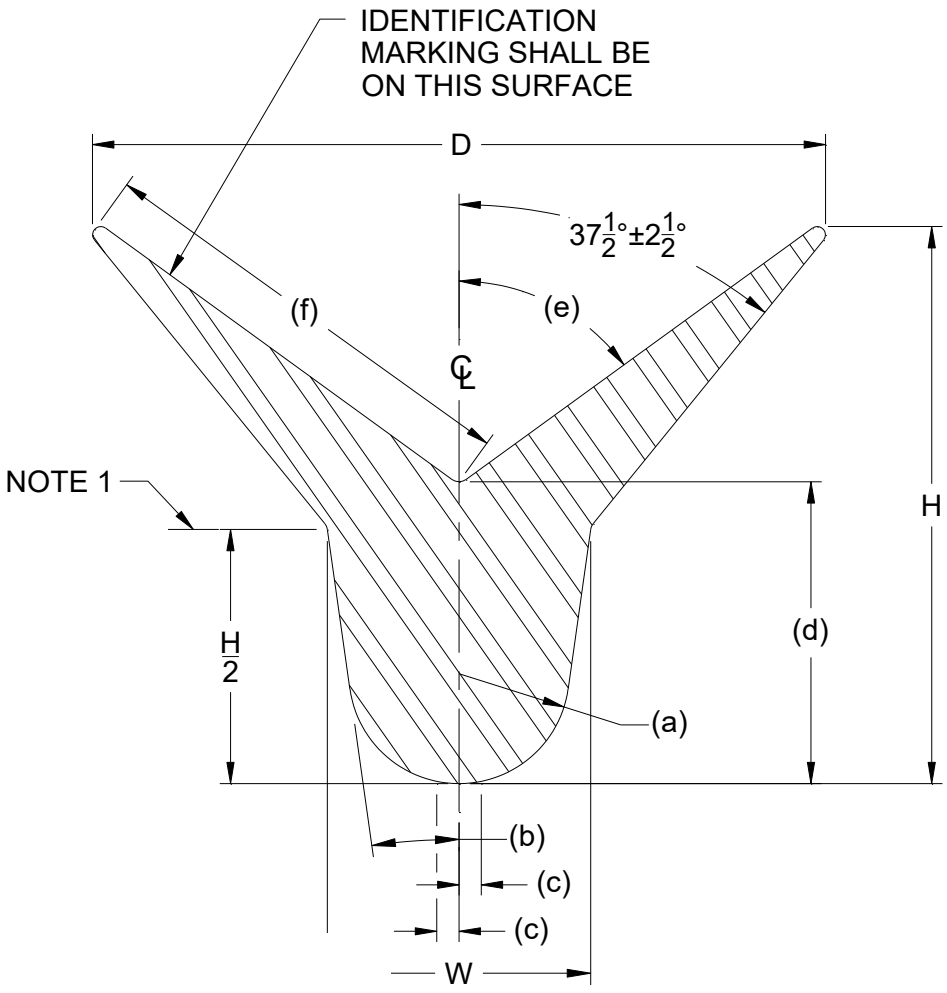


FIGURE 2. Class 3 solid ring inserts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

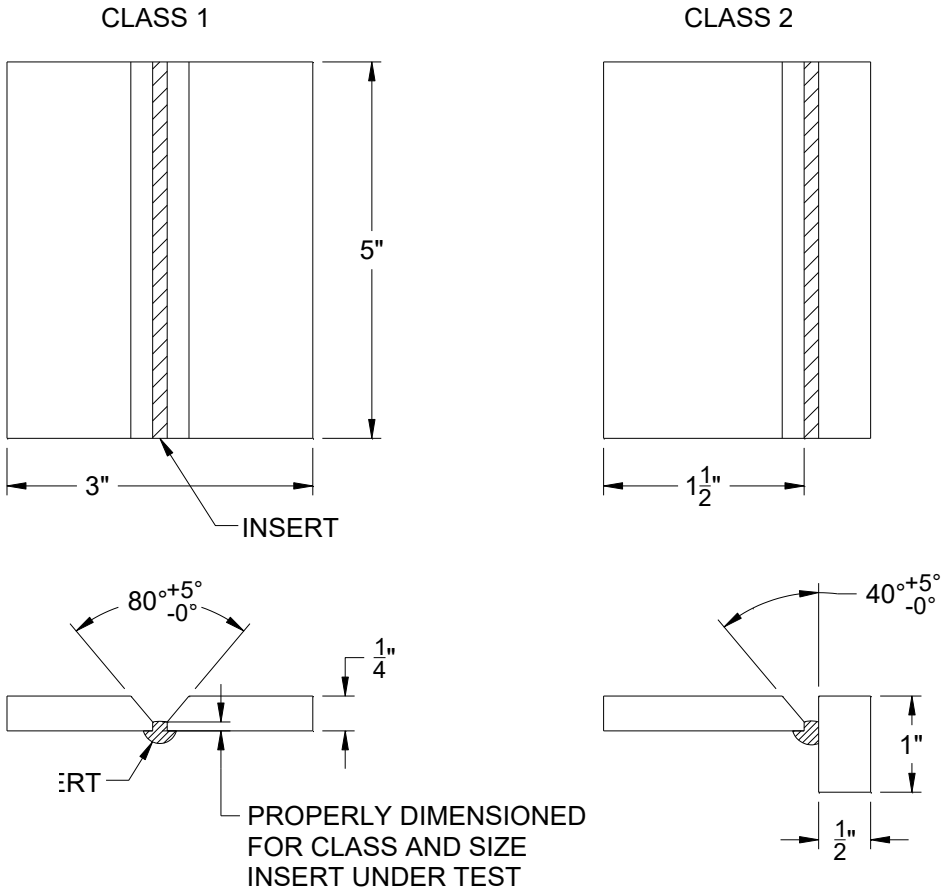


NOTES:

1. Reference diameter for correlating with pipe I.D.
2. For dimensions and tolerances, see table VIII.
3. When specified, rings 1½ to 2 inches in diameter shall be formed of 5/32-inch material.

FIGURE 3. Class 4 inserts – cross-sectional configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

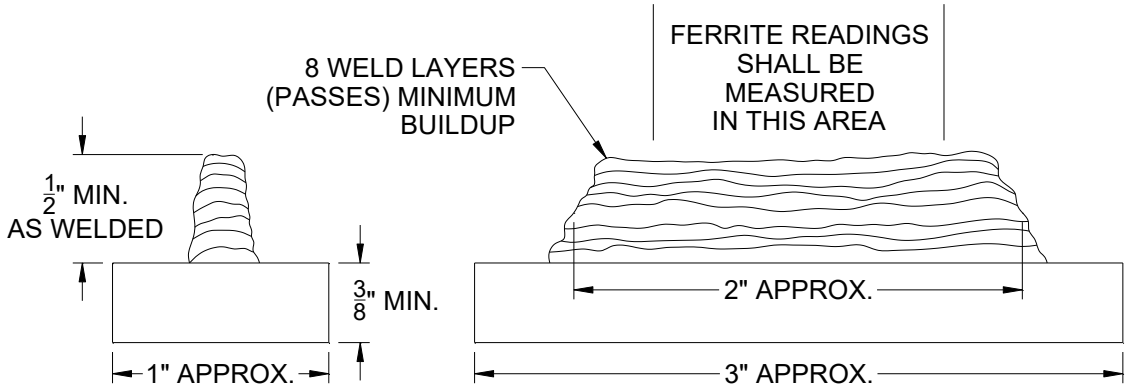


NOTES:

1. The base plate shall be of a composition similar to the insert.
2. One-half of the joint shall be welded in the flat position and the remainder in the vertical position. All welding shall be done on the groove side.
3. Argon gas shall be used for purging and shielding the electrode and the back side of the weld.
4. Standard techniques and procedures for fusing the insert shall be followed at all times.

FIGURE 4. Fusibility test joint.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.



NOTES:

1. The base material for the weld pad shall be type 301, 302, or 304 conforming to ASTM A240/A240M.
2. The weld pad shall be built up by depositing single bead layers one on top of the other to a minimum height of ½ inch. The weld bead layers shall be deposited with a maximum width in accordance with the table below, except the first several layers may be deposited with a wider weave.
3. The argon gas tungsten arc welding process shall be used. The weld stops and starts shall be located at the ends of the weld buildup.
4. The weld pad shall be cooled between passes by water quenching immediately after each pass, except the last two weld passes shall be air cooled below 800 °F prior to water quenching.
5. Each weld pass shall be cleaned prior to depositing the next weld bead.
6. After welding, the weld buildup shall be surface ground to 64 root mean square (rms) finish to provide sufficient finished surface to make the required ferrite readings. The ground area need not be continuous along the specimen. The weld buildup shall be continuously water cooled during the grinding operation. The finished ground surface shall be approximately parallel with the plate surface and shall be located within the last two passes or layers of weld buildup. Machining and metallographic polishing sufficient to remove the cold working produced by machining may be used in lieu of surface grinding. The weld buildup should be water cooled during the machining operation.
7. A total of six ferrite readings shall be taken on the machined surface along the longitudinal axis of the weld bead. The readings should not be taken on either end where the weld craters (stop and start areas) are located. Readings should be taken in areas having dimensions at least 1¼ times the diameter of the magnet. Ferrite readings shall be taken by the Magne-Gage, or equal.
8. The six readings obtained shall be averaged to a single value for conversion to FN.

Insert			**Weld bead**
Class	**Size**	**Style**	**Layer width – inch maximum**
1	⅛	--	¼
1	5/32	--	5/16
2	⅛	--	¼
2	5/32	--	5/16
3	--	D	¼
3	--	E	5/16
4	⅛	--	¼
4	5/32	--	⅜

FIGURE 5. Ferrite weld specimen.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 3439-2025-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T16:40Z
Check the source to verify that this is the current version before use.